

Topic 5: Arms Race

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Overview



Scenario: Purple (P) and Green (G) are two countries competing in an arms race.

Research Question: What happens to the long-term spending of each country?

Key Factors:

- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises
- Does public sentiment affect defence spending?
- Does limiting spending affect a country's position?

Motivation: It helps predict military escalation between countries, which can inform policies to promote peace and prevent conflict.

Base Model



$$\frac{\partial p}{\partial t} = ag(t) - bp(t)$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t)$$

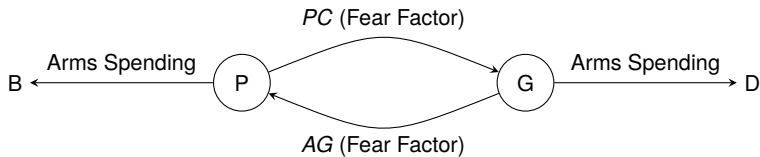
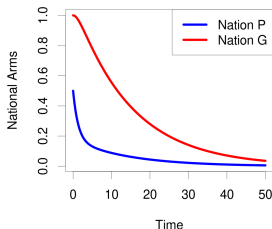


Figure: Base Model

Long Term Behavior (Base Model)

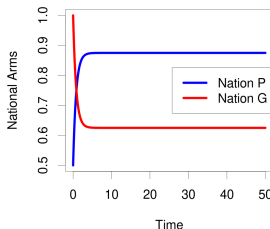
$$bd > ac$$

$$a=0.1, b=0.7, c=0.2, d=0.1$$



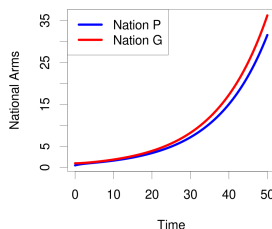
$$bd = ac$$

$$a=0.7, b=0.5, c=0.5, d=0.7$$



$$bd < ac$$

$$a=0.5, b=0.5, c=0.2, d=0.1$$



Complete mutual
disarmament
 $(p^\infty, g^\infty) = (0, 0)$

Stabilized spending
 $(p^\infty, g^\infty) =$
 $(\frac{cp_0 + bg_0}{a+c}, \frac{dp_0 + ag_0}{a+c})$

Perpetual arms race
 $(p^\infty, g^\infty) = (\infty, \infty)$

Public Sentiment (Trust and Hostility)



$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$

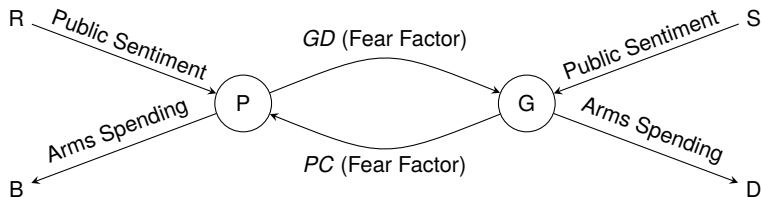


Figure: Base Model Including Public Sentiment

Long Term Behavior (Public Sentiment)



- provide plots
- provide stability conditions and convergences
- This model will take on negative values so discuss that by either capping the value or other methods



Mutual Logistic Fear

Limitation of Current Model: Does not account for the limited resources each country has, so introduce self limiting logistic growth

$$\frac{\partial p}{\partial t} = a\left(1 - \frac{p(t)}{K_p}\right)g(t) - bp(t) + r$$
$$\frac{\partial g}{\partial t} = c\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + s$$

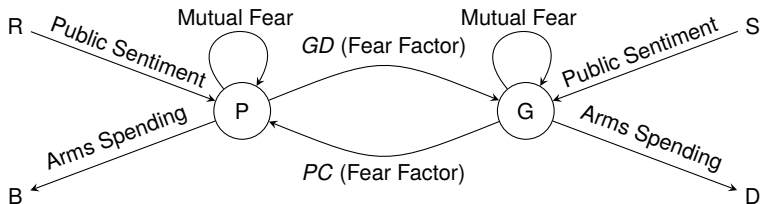


Figure: Base Model Including Public Sentiment and Mutual Fear

Long Term Behavior (Mutual Logistic Fear)



- provide plots
- provide stability conditions
- provide convergences (if possible)

Thank you for listening!



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